

CSL-351: Computer Networks



Lab - 10

Cisco Packet Tracer

Dr. Anand M. Baswade
Indian Institute of Technology, Bhilai
anand@iitbhilai.ac.in

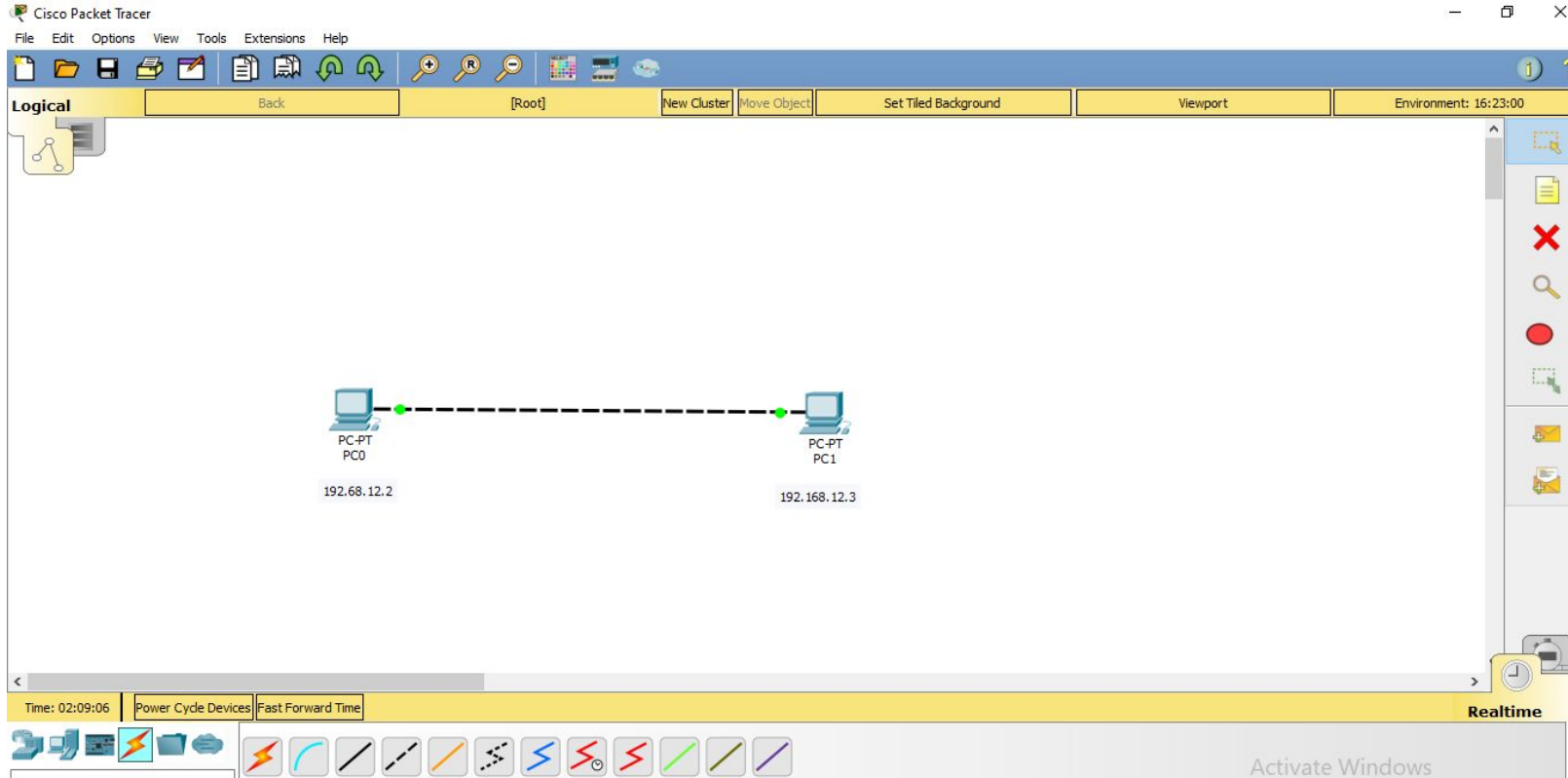
Content

- ❑ Introduction to Cisco Packet Tracer
- ❑ Installation
- ❑ Assign IP's in CPT
- ❑ Connection with Hub
- ❑ Connection with Switch
- ❑ Creation of Multiple LAN's
- ❑ Creation of WAN's
- ❑ DHCP

Introduction

- Cisco Packet Tracer is a network simulation software developed by Cisco Systems
- **Learning network concepts:** Packet Tracer can help you learn about the fundamentals of networking, such as the OSI model, TCP/IP, subnetting, and routing.
- **Designing networks:** You can use Packet Tracer to design new networks or to improve existing networks.
- **Troubleshooting networks:** Packet Tracer can help you identify and troubleshoot network problems in a safe virtual environment.

How to level the IP address



Steps

- Allocate IP addresses to the PCs

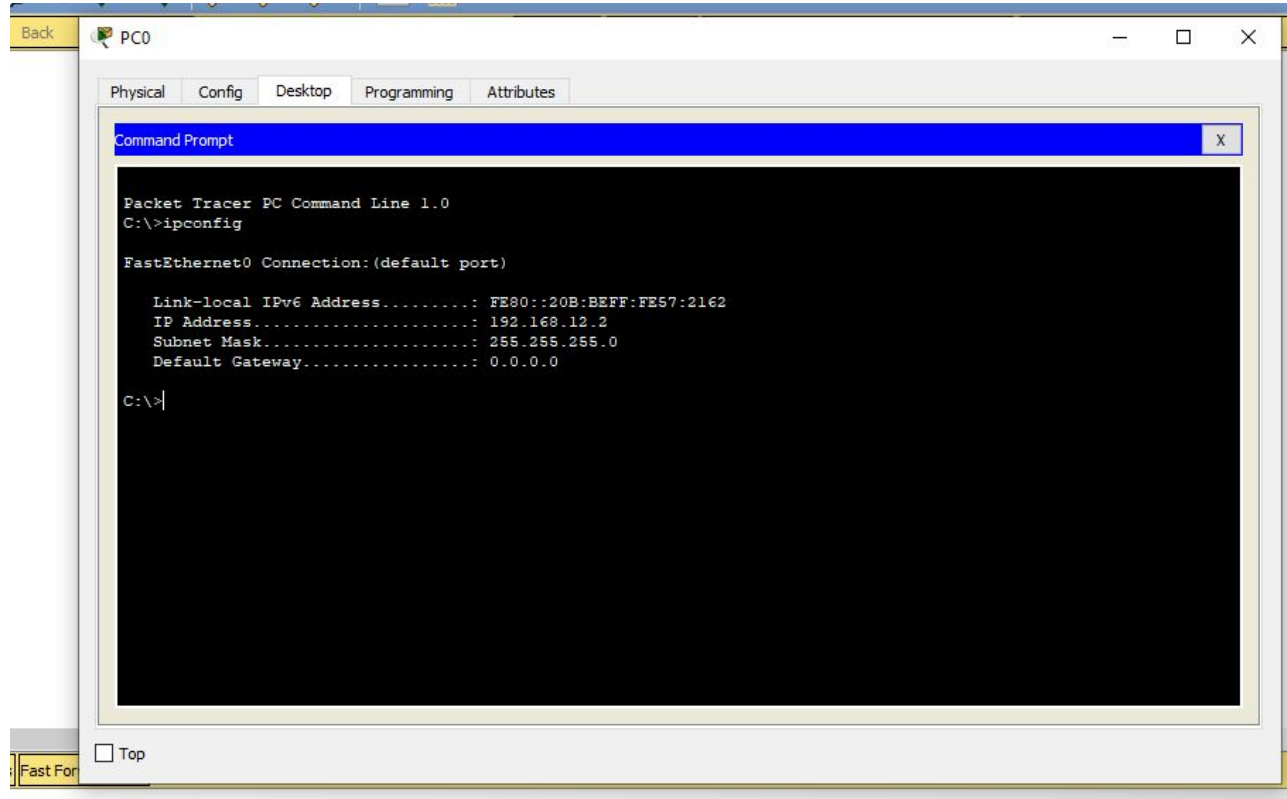
PC0

The screenshot shows the configuration window for PC0. The 'IP Configuration' tab is active. Under 'IP Configuration', the 'Static' radio button is selected. The 'IP Address' field is set to 192.168.12.2, 'Subnet Mask' is 255.255.255.0, 'Default Gateway' is 0.0.0.0, and 'DNS Server' is 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is FE80::20B:8EFF:FE57:2162, 'IPv6 Gateway' is empty, and 'IPv6 DNS Server' is empty. The window has a 'Back' button and a 'Top' checkbox at the bottom left.

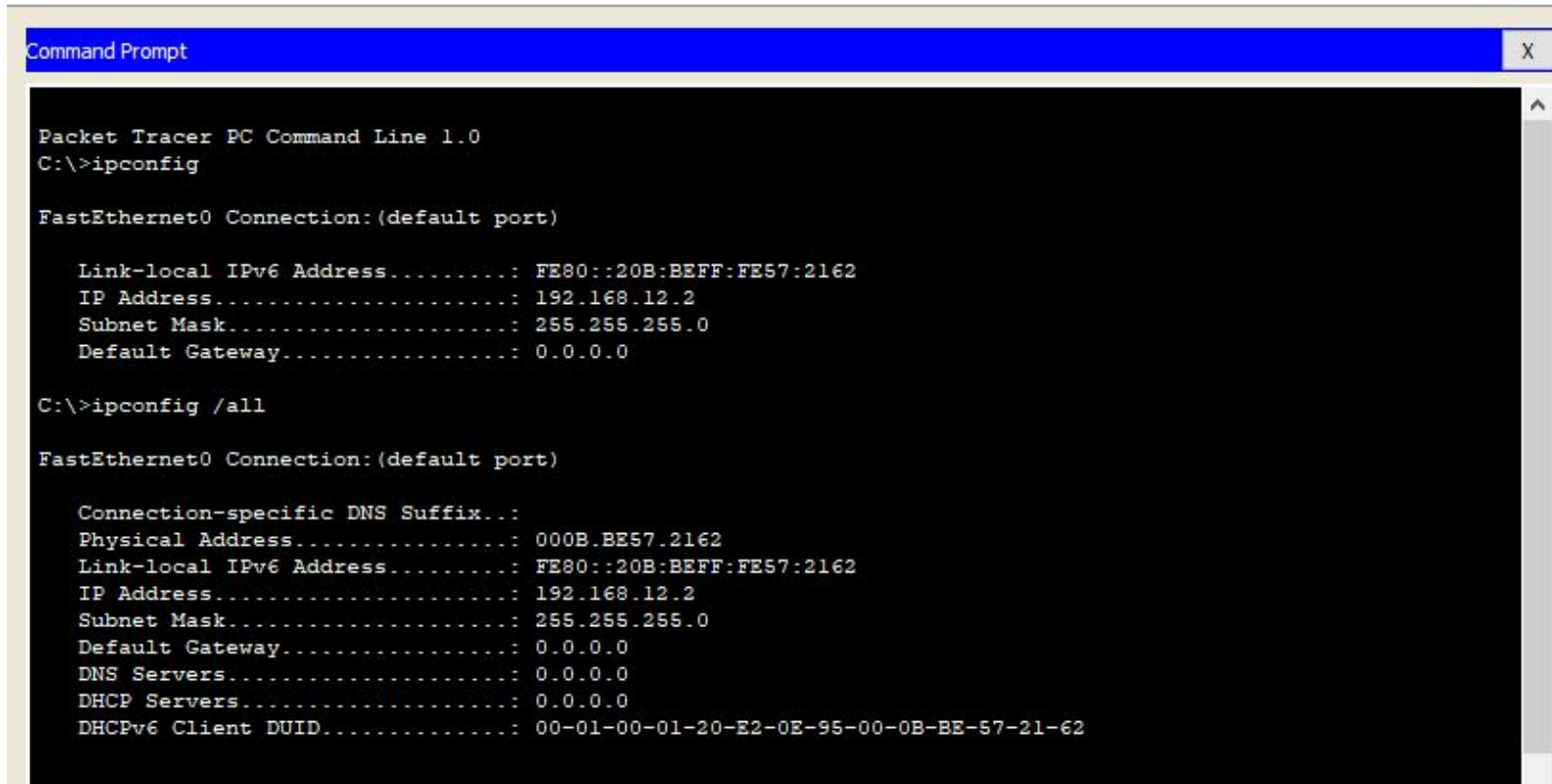
PC1

The screenshot shows the configuration window for PC1. The 'IP Configuration' tab is active. Under 'IP Configuration', the 'Static' radio button is selected. The 'IP Address' field is set to 192.168.12.3, 'Subnet Mask' is 255.255.255.0, 'Default Gateway' is 0.0.0.0, and 'DNS Server' is 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is FE80::260:3EFD:FEDD:4642, 'IPv6 Gateway' is empty, and 'IPv6 DNS Server' is empty. The window has a 'Back' button and a 'Top' checkbox at the bottom left.

To check the ip address of PC0



To check the MAC address



```
Command Prompt X
Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

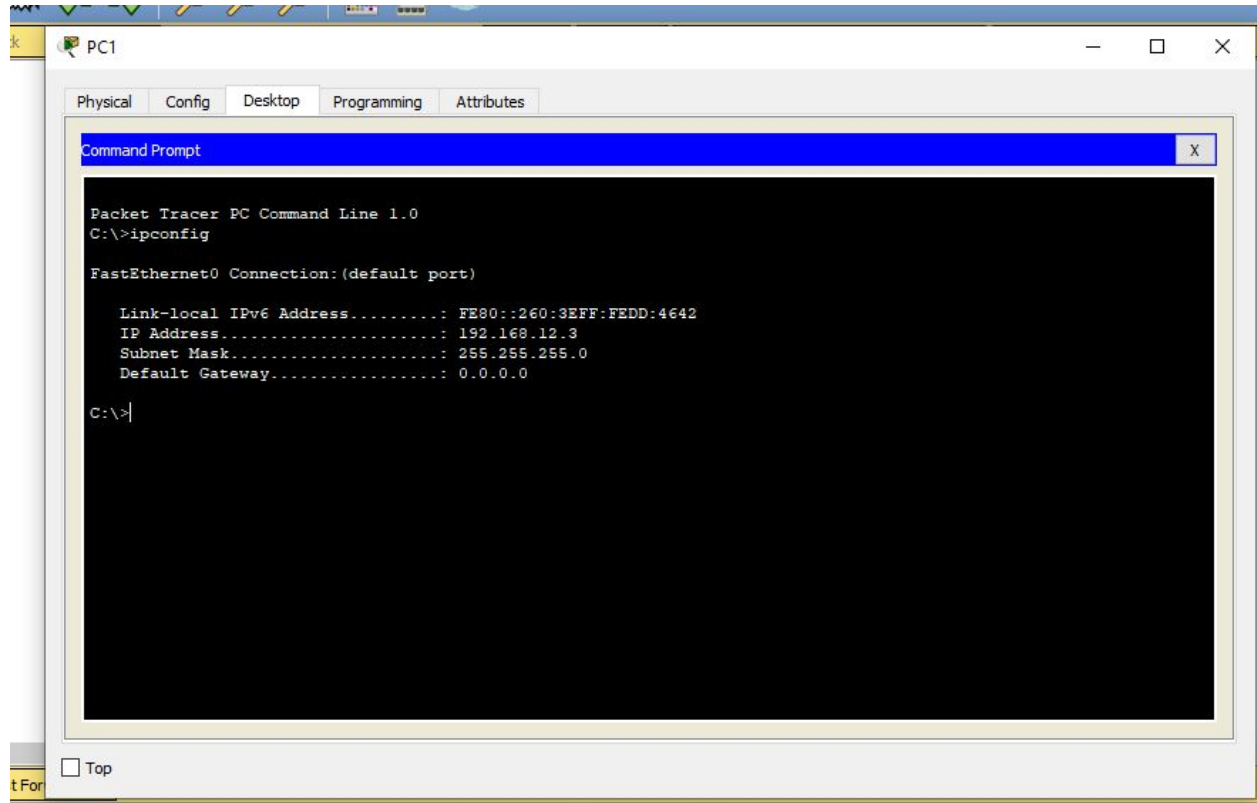
    Link-local IPv6 Address . . . . . : FE80::20B:BEFF:FE57:2162
    IP Address. . . . . : 192.168.12.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 0.0.0.0

C:\>ipconfig /all

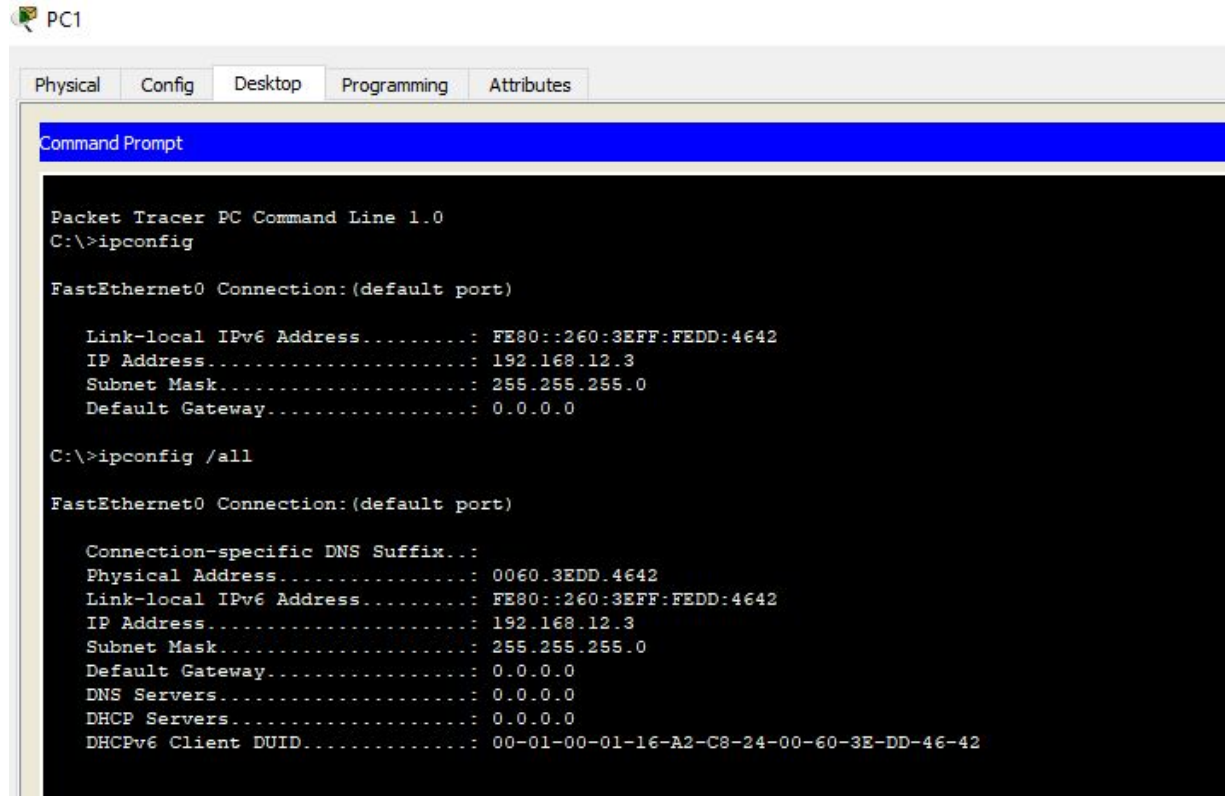
FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix... :
    Physical Address. . . . . : 000B.BE57.2162
    Link-local IPv6 Address . . . . . : FE80::20B:BEFF:FE57:2162
    IP Address. . . . . : 192.168.12.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 0.0.0.0
    DNS Servers . . . . . : 0.0.0.0
    DHCP Servers . . . . . : 0.0.0.0
    DHCPv6 Client DUID. . . . . : 00-01-00-01-20-E2-0E-95-00-0B-BE-57-21-62
```

To check the ip address of PC1



To check the MAC address of PC1



The screenshot shows a Packet Tracer PC1 interface with a Command Prompt window open. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt displays the output of the 'ipconfig' and 'ipconfig /all' commands.

```
Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Link-local IPv6 Address . . . . . : FE80::260:3EFF:FEDD:4642
    IP Address. . . . . : 192.168.12.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 0.0.0.0

C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix... : 
    Physical Address. . . . . : 0060.3EDD.4642
    Link-local IPv6 Address . . . . . : FE80::260:3EFF:FEDD:4642
    IP Address. . . . . : 192.168.12.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 0.0.0.0
    DNS Servers . . . . . : 0.0.0.0
    DHCP Servers . . . . . : 0.0.0.0
    DHCPv6 Client DUID. . . . . : 00-01-00-01-16-A2-C8-24-00-60-3E-DD-46-42
```

Connection of switch to form LAN

- Connect End Devices to the ports of Switch
- Use **copper straight through** cable to connect
- Give ip addresses to all the individual pc, level the ip address for each pc
- Check using ping command
- Go to simulation send the packet

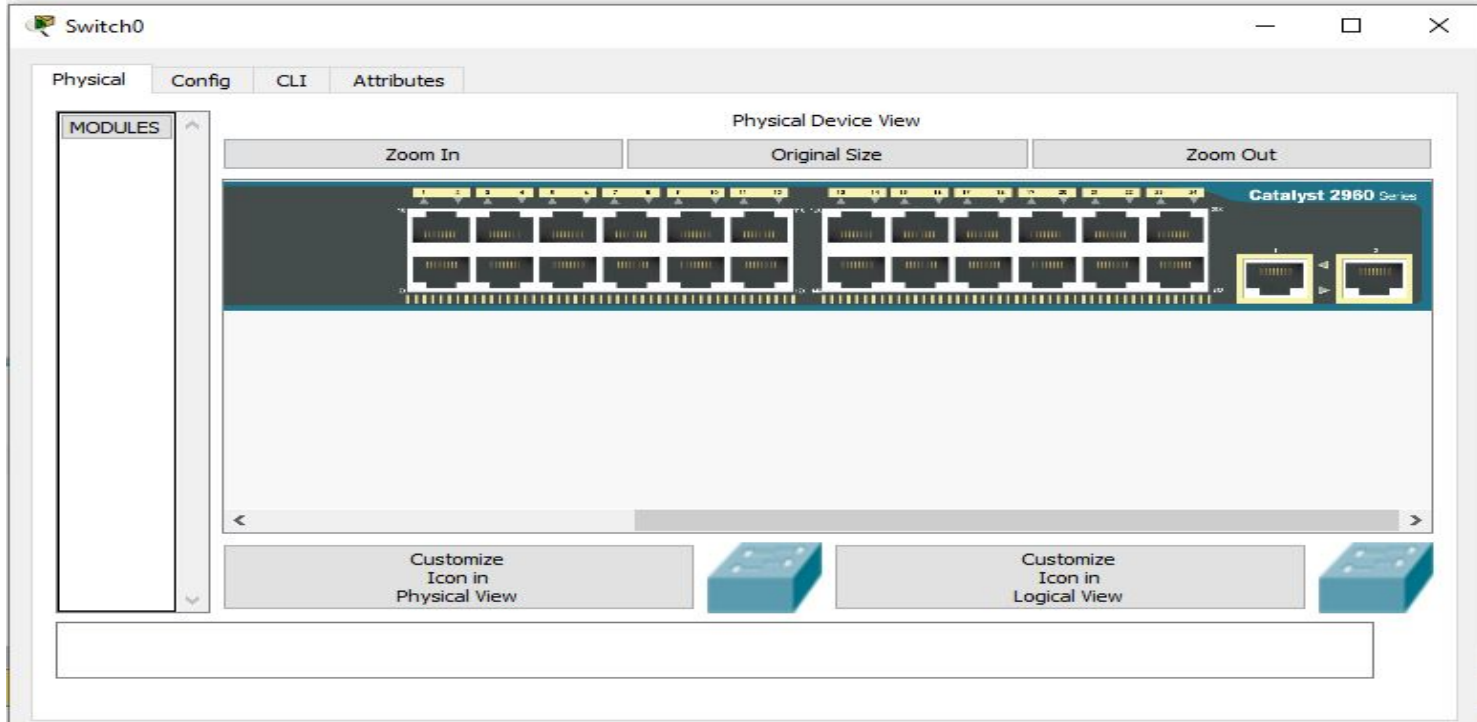
Connection of switch



Creation of LAN using switch

- Create single local area network
- Required components
 - 3 end devices
 - 1 switch

Physical view of switch



24 Fast Ethernet port and 2 Gigabit Ethernet port



Logical

Back

[Root]

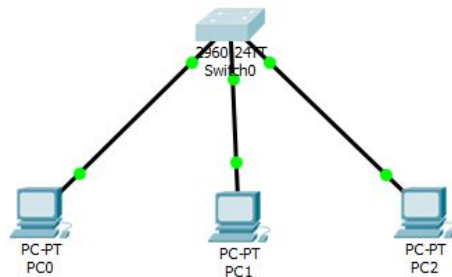
New Cluster

Move Object

Set Tiled Background

Viewport

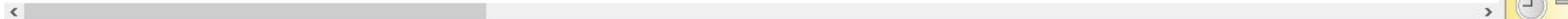
Environment: 04:51:30



192.168.10.1

192.168.10.2

192.168.0.3



Time: 00:11:06

Power Cycle Devices

Fast Forward Time



Scenario 0

New

Delete

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	PC1	ICMP		0.000	N	0

Activate Windows

Go to Settings to activate Windows.

MAC Address Tables

- A MAC address table, sometimes called a Content Addressable Memory (CAM) table, is used on Ethernet switches to determine where to forward traffic on a LAN.
- Now let's break this down a little bit to understand how the MAC address table is built and used by an Ethernet switch to help traffic move along the path to its destination.

IOS Command Line Interface

changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
changed state to up

Switch>EN

Switch#show mac-address-table

Mac Address Table

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
1	0060.4700.ae4e	DYNAMIC	Fa0/1
1	00d0.5850.0467	DYNAMIC	Fa0/2
1	00e0.b09c.4ac6	DYNAMIC	Fa0/3

Switch#

Ctrl+F6 to exit CLI focus

Copy

Paste

Creation of Multiple LAN

- Create two local area network and connect via router
- Required components
 - 6 end devices
 - 2 switches (2960)
 - 1 router (2911)

Steps

- Place all the devices in the workspace
- Assign IP addresses to all the PCs of LAN1

10.0.0.2, 10.0.0.3, 10.0.0.4

- Assign IP addresses to all the PCs of LAN2

192.168.0.2, 192.168.0.3, 192.168.0.4

Set the default gateway for each end device

PC0

Physical Config Desktop Programming Attributes

IP Configuration

☐ DHCP ☒ Static

IP Address: 10.0.0.2

Subnet Mask: 255.0.0.0

Default Gateway: 10.0.0.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::20D:BDFF:FE58:D2EC

IPv6 Gateway:

IPv6 DNS Server:

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/0

Port Status

☒ On

Bandwidth

☐

1000 Mbps

☒

100 Mbps

☐

10 Mbps

☒

Auto

Duplex

☐

Half Duplex

☒

Full Duplex

☒

Auto

MAC Address

00E0.8FCC.BE01

IP Configuration

IP Address

10.0.0.1

Subnet Mask

255.0.0.0

Tx Ring Limit

10

Equivalent IOS Commands

```
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```



Logical

Back

[Root]

New Cluster

Move Object

Set Tiled Background

Viewport

Environment: 15:07:00

LAN1: 10.0.0.0
SubnetMask :255.0.0.0

2960-24T
Switch0



10.0.0.2



10.0.0.3



10.0.0.4



2911
Router0

2960-24T
Switch1

LAN2 : 192.168.0.0
Subnet Mask: 255.255.255.0



192.168.0.2



192.168.0.3



192.168.0.4

Time: 00:33:22

Power Cycle Devices

Fast Forward Time



Scenario 0

New

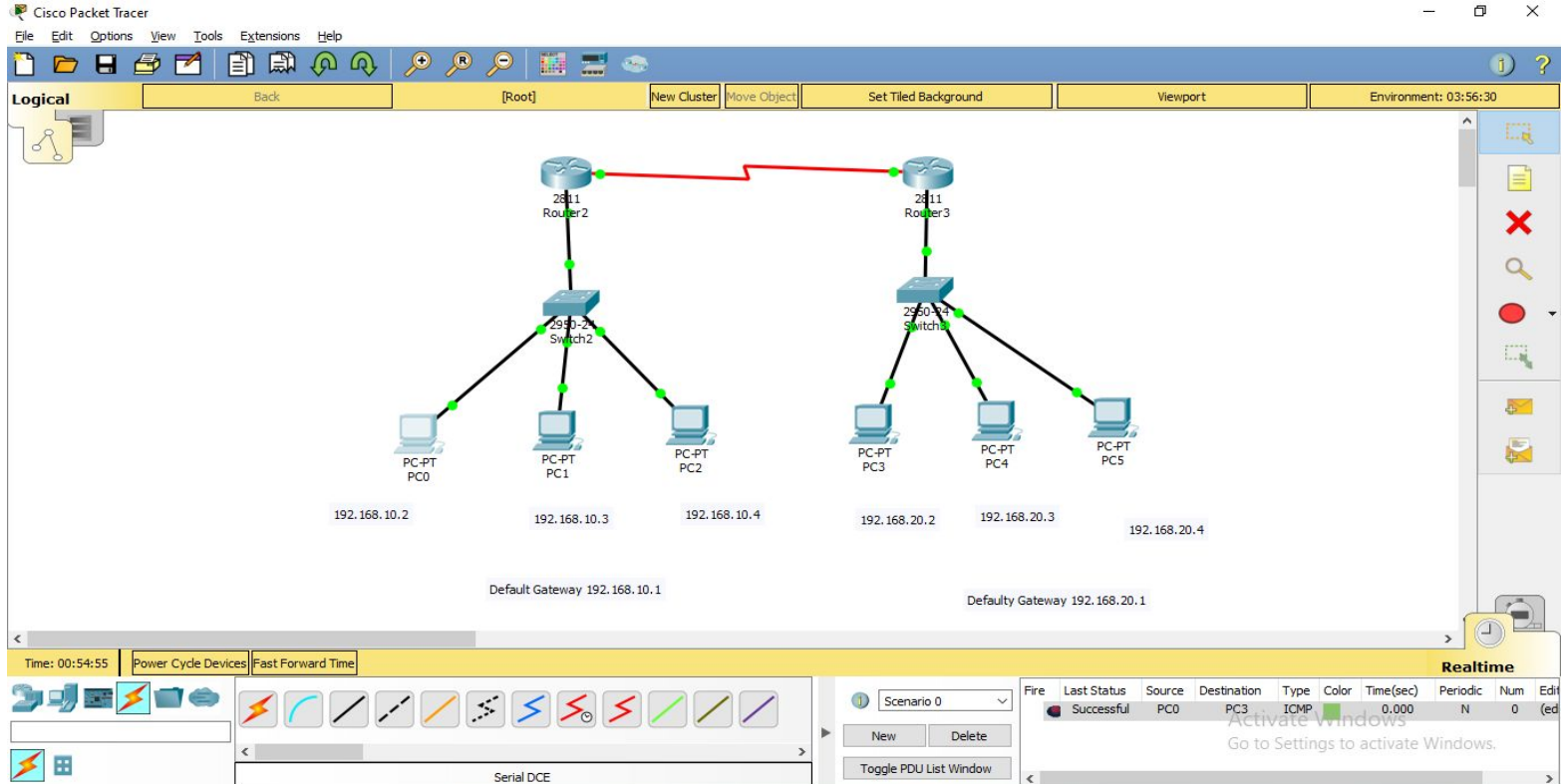
Delete

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	E
Successful		PC0	Router0	ICMP		0.000	N	0	(
Successful		PC0	PC3	ICMP		0.000	N	1	(

Realtime

Go to Settings to activate Windows.

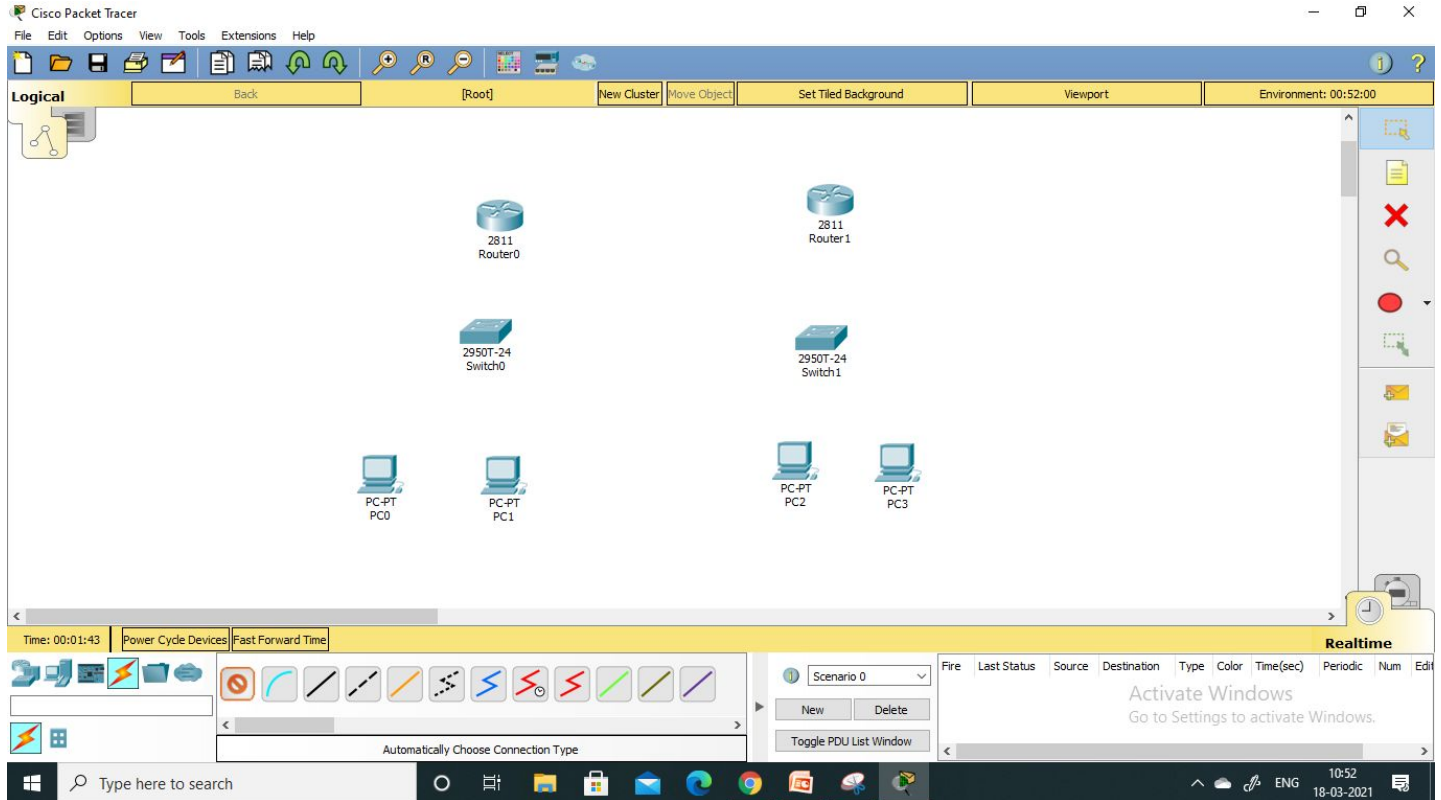
Creation of WAN

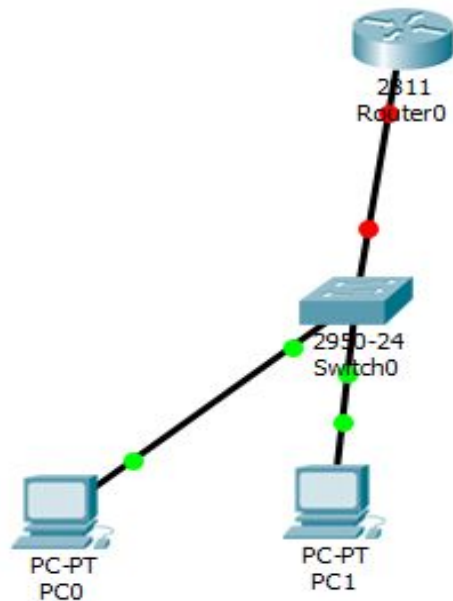


Creation of Network

- Create single network
- Required components
 - 6 end devices
 - 2 switch (2950-24T)
 - 2 Router (2811)

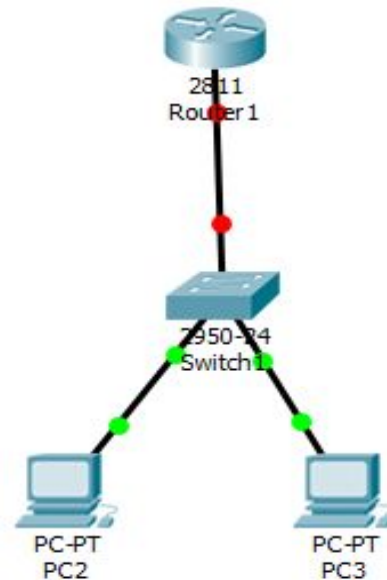
Place the devices





192.168.10.2

192.168.10.3



192.168.20.2

192.168.20.3

- Assign IP to all the pc and level it
- Give the gateway address all for both the LANs

PC0

Physical Config Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IP Address

192.168.10.2

Subnet Mask

255.255.255.0

Default Gateway

19.168.10.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ DHCP

☐ Auto Config

☒ Static

IPv6 Address

 /

Link Local Address

FE80::20D:BDFF:FE84:2A0D

IPv6 Gateway

IPv6 DNS Server

Top

Physical Config Desktop Programming Attributes

IP Configuration

X

IP Configuration

☐ DHCP☒ Static

IP Address

192.168.20.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.20.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ DHCP☐ Auto Config☒ Static

IPv6 Address

Link Local Address

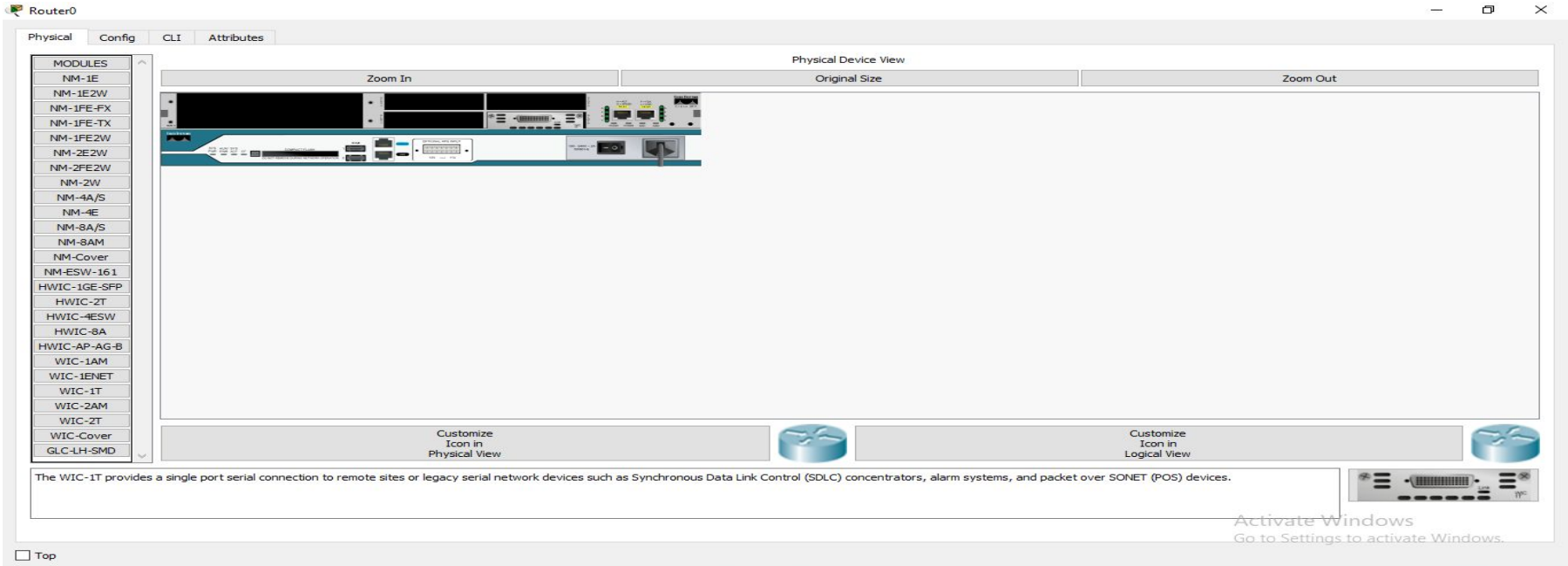
FE80::290:2BFF:FE43:4BB7

IPv6 Gateway

IPv6 DNS Server

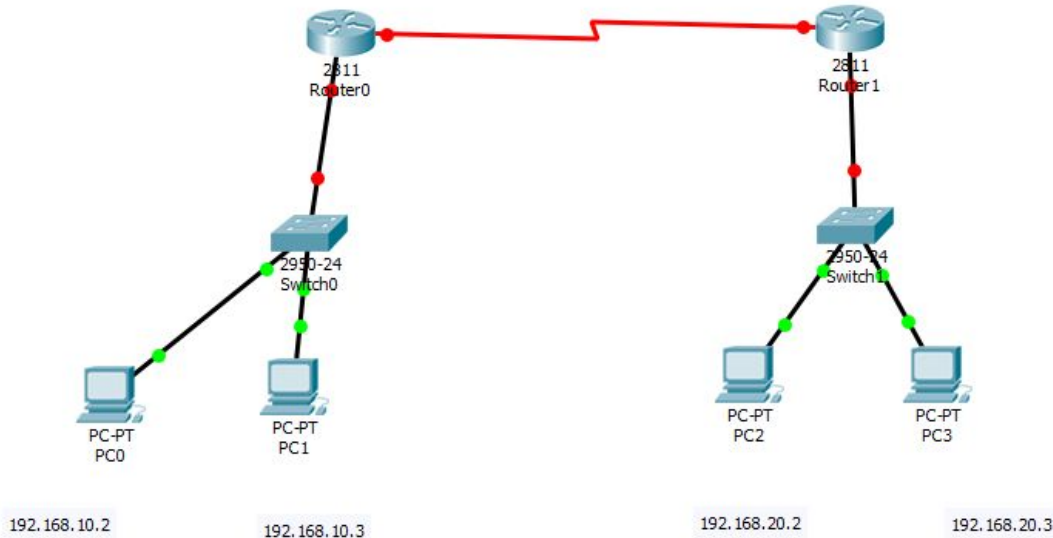
Steps

- Go the 1st router: in physical tab, switch off
- Drag and drop WIC-1T and switch on the power
- Similarly do it for the second router



Steps

- Use Serial DCE for connection between two routers
 - Drag and drop WIC-1T and switch on the power
 - Similarly do it for the second router



Go the router

The image shows a network diagram on the left and a configuration window for Router0 on the right. The network diagram includes a 2811 Router0 connected to a 2850T-24 Switch0, which is connected to two PCs (PC0 and PC1). A red line highlights the connection between the router and the switch. The configuration window for Router0 shows the 'Config' tab with a tree view on the left and configuration options on the right. The 'Serial0/0/0' interface is selected in the tree view. The configuration options for Serial0/0/0 include Port Status, Duplex (Full Duplex), Clock Rate (2000000), IP Configuration, IP Address, Subnet Mask, and Tx Ring Limit (10). Below the configuration options, there is a section for 'Equivalent IOS Commands'.

10.10.1.3

Router0

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- FastEthernet0/0
- FastEthernet0/1
- Serial0/0/0

Serial0/0/0

Port Status

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IP Address

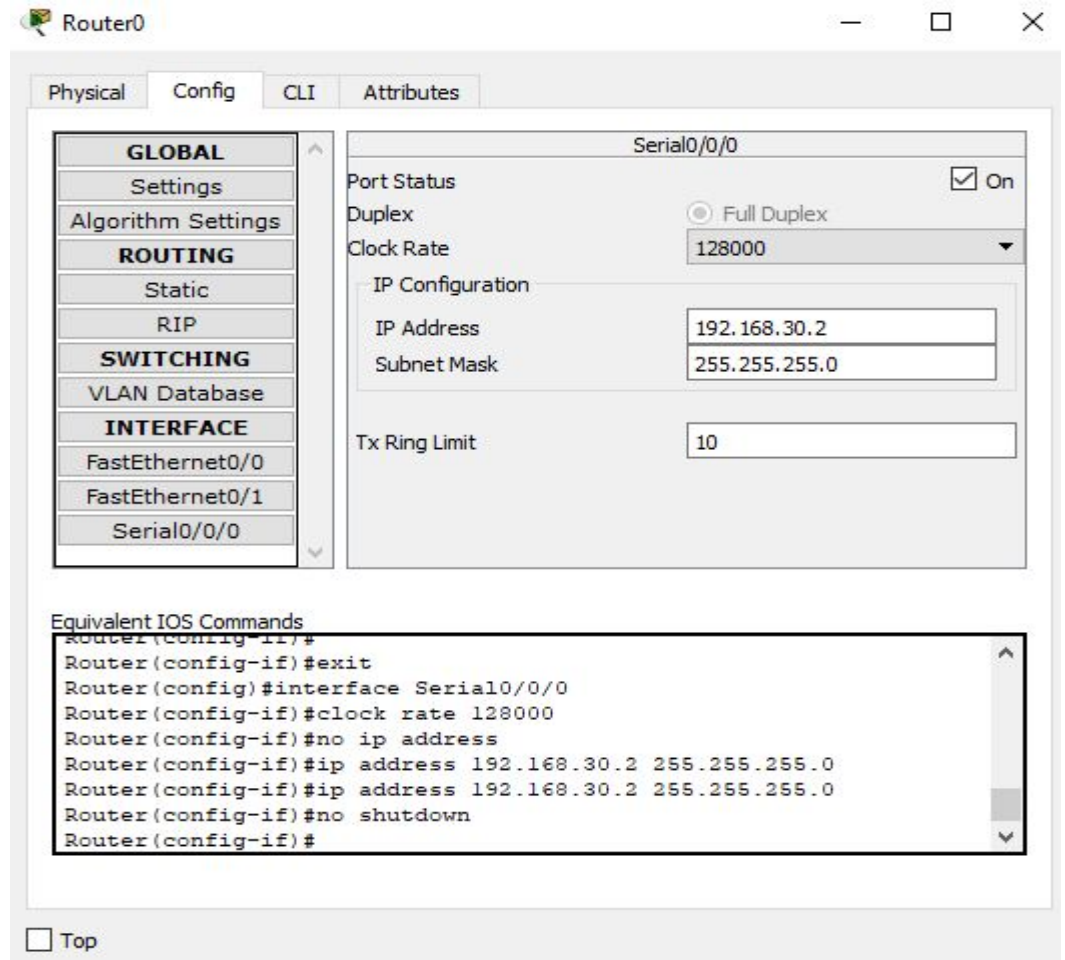
Subnet Mask

Tx Ring Limit 10

Equivalent IOS Commands

```
Enter configuration commands, one per line. End with CNTRL-Z
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#
```

- Router > config > serial 0/0/0
- In router, clock rate is to be given 128000
- In router, give the ip
- After setting all values, set Port status ON.



The screenshot shows the configuration window for Router0 in Cisco Packet Tracer. The window has tabs for Physical, Config, CLI, and Attributes. The Config tab is active, showing a tree view on the left with categories: GLOBAL (Settings, Algorithm Settings), ROUTING (Static, RIP), SWITCHING (VLAN Database), and INTERFACE (FastEthernet0/0, FastEthernet0/1, Serial0/0/0). The Serial0/0/0 interface is selected, and its configuration is shown on the right. The Port Status is checked (On), Duplex is set to Full Duplex, and Clock Rate is set to 128000. The IP Configuration section shows IP Address 192.168.30.2 and Subnet Mask 255.255.255.0. The Tx Ring Limit is set to 10. Below the configuration, the Equivalent IOS Commands are listed in a text box.

Router0

Physical Config CLI Attributes

Serial0/0/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 128000

IP Configuration

IP Address 192.168.30.2

Subnet Mask 255.255.255.0

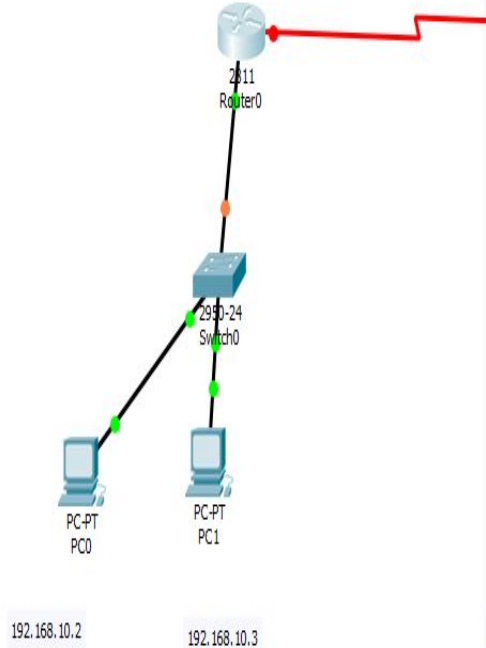
Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface Serial0/0/0  
Router(config-if)#clock rate 128000  
Router(config-if)#no ip address  
Router(config-if)#ip address 192.168.30.2 255.255.255.0  
Router(config-if)#ip address 192.168.30.2 255.255.255.0  
Router(config-if)#no shutdown  
Router(config-if)#
```

☐ Top

- Router > config > FastEthernet 0/0
- In router, give the IP which is same as default gateway 192.168.10.1
- Turn it on



Router0

Back [Root] New Cluster Move

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Serial0/0/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0002.1780.0201

IP Configuration

IP Address 192.168.10.1

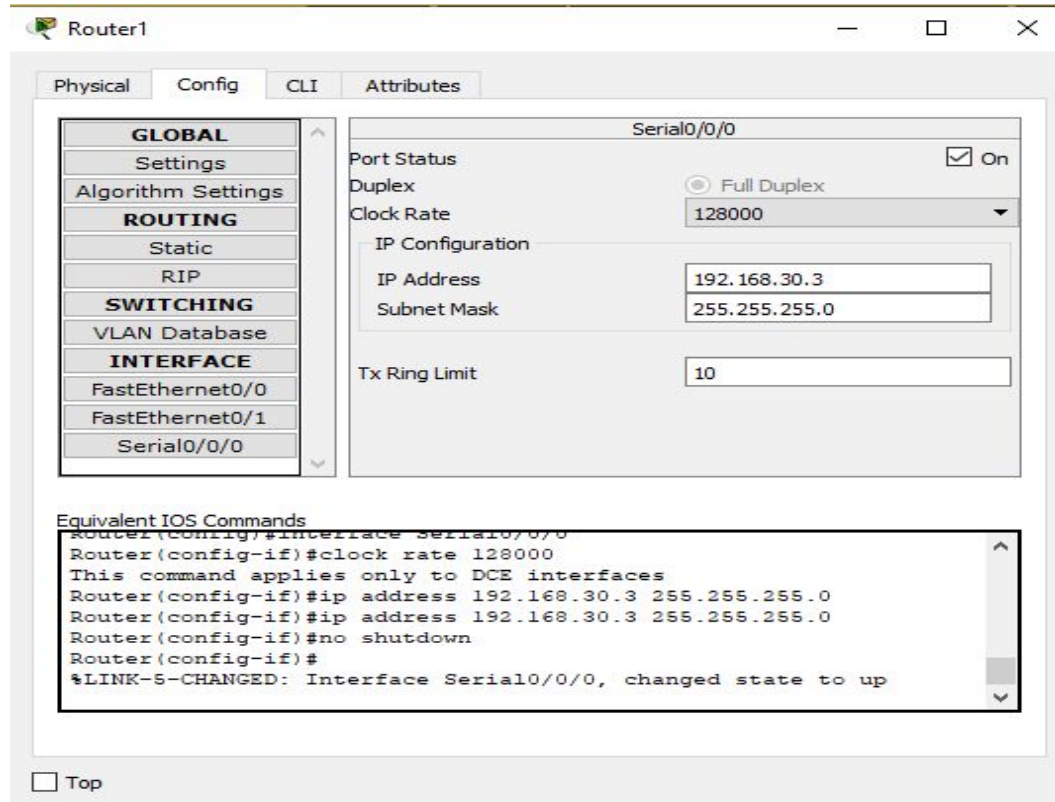
Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
```


- Repeat the same procedure for the second router, Remember the Interfaces it has been connected to



IOS Command Line Interface

```
Router(config-if)#  
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to  
administratively down  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0,  
changed state to down  
  
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up  
shutdown  
Router(config-if)#  
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to  
administratively down  
no shutdown  
Router(config-if)#  
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0,  
changed state to up  
  
Router(config-if)#exit  
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.30.1  
Router(config)#exit  
Router#  
%SYS-5-CONFIG_I: Configured from console by console
```

Ctrl+F6 to exit CLI focus

Copy

Paste

IOS Command Line Interface

```
Router(config-if)#ip address 192.168.30.2 255.255.255.0
Router(config-if)#ip address 192.168.30.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0,
changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

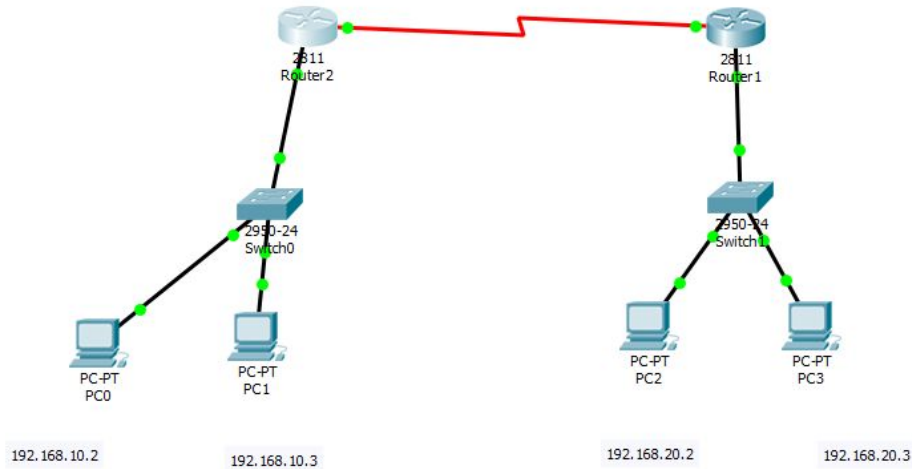
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.30.1
Router(config)#exit
Router#
%SYS-5-CONFIG-I: Configured from console by console
```

Ctrl+F6 to exit CLI focus

Copy

Paste

Packet transmission



Over Cycle Devices Fast Forward Time

Serial DCE

Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	PC2	ICMP		0.000	N	0

Realtime

Activate Windows
Go to Settings to activate Windows.

